

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 09

Sequential Circuit



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Recap of Previous Lecture



Synch. Counter

Question Discussion





Topics to be Covered

Ring Counter

Johnson Counter

Shift register

[Ring Counter]

4-bit Ring Counter :

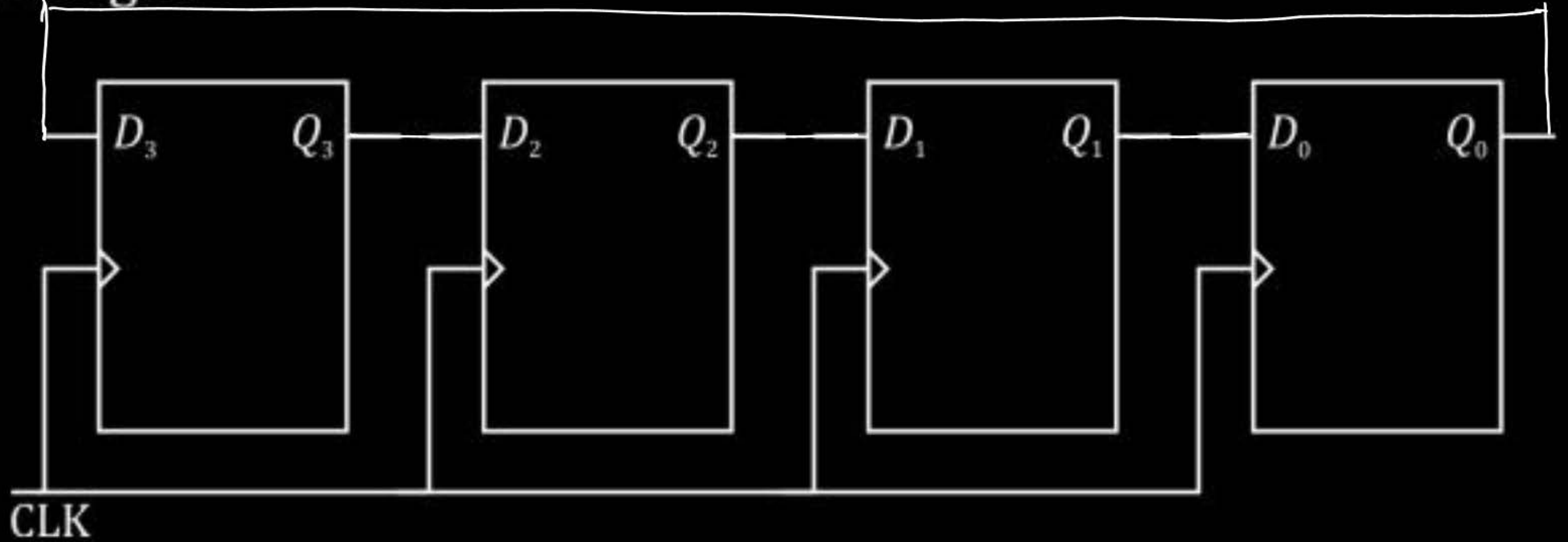
Circuit Design :

$$Q_3(n+1) = D_3 = Q_0(n)$$

$$Q_0(n+1) = D_0 = Q_1(n)$$

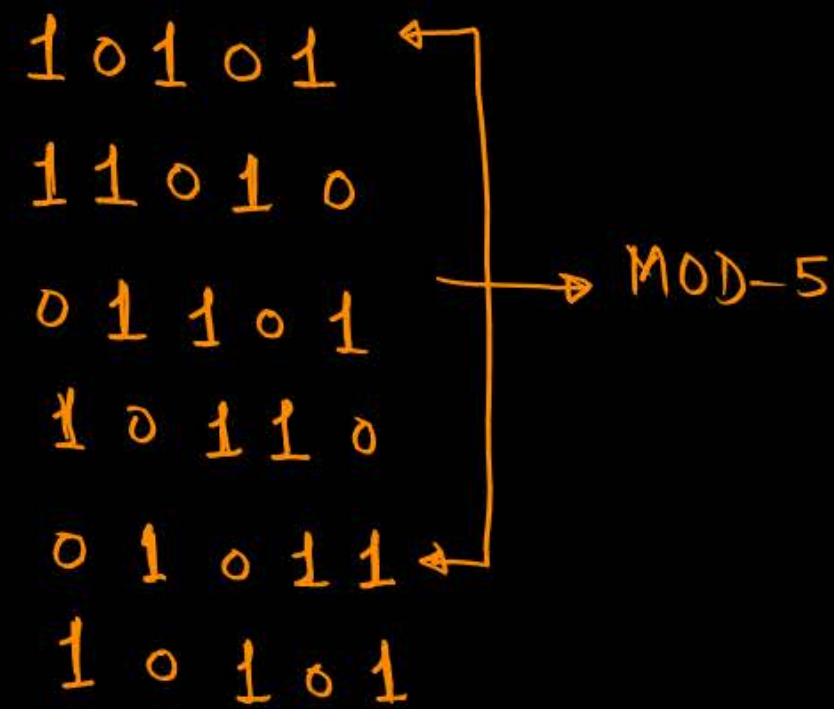
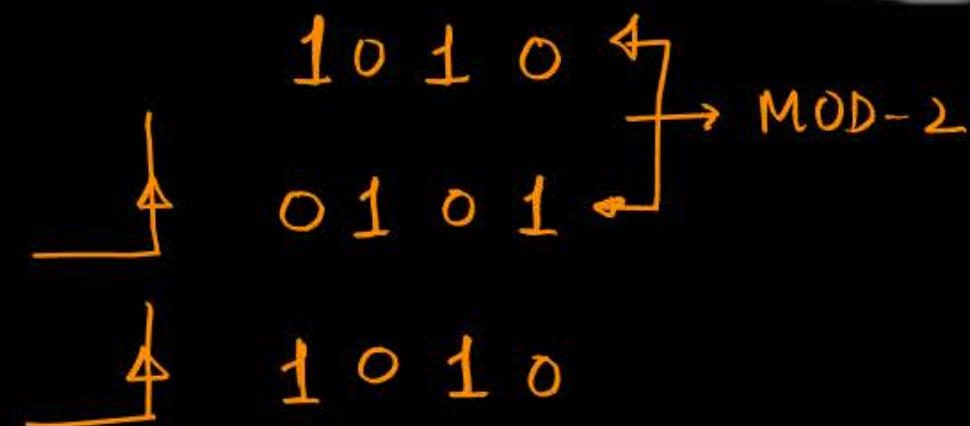
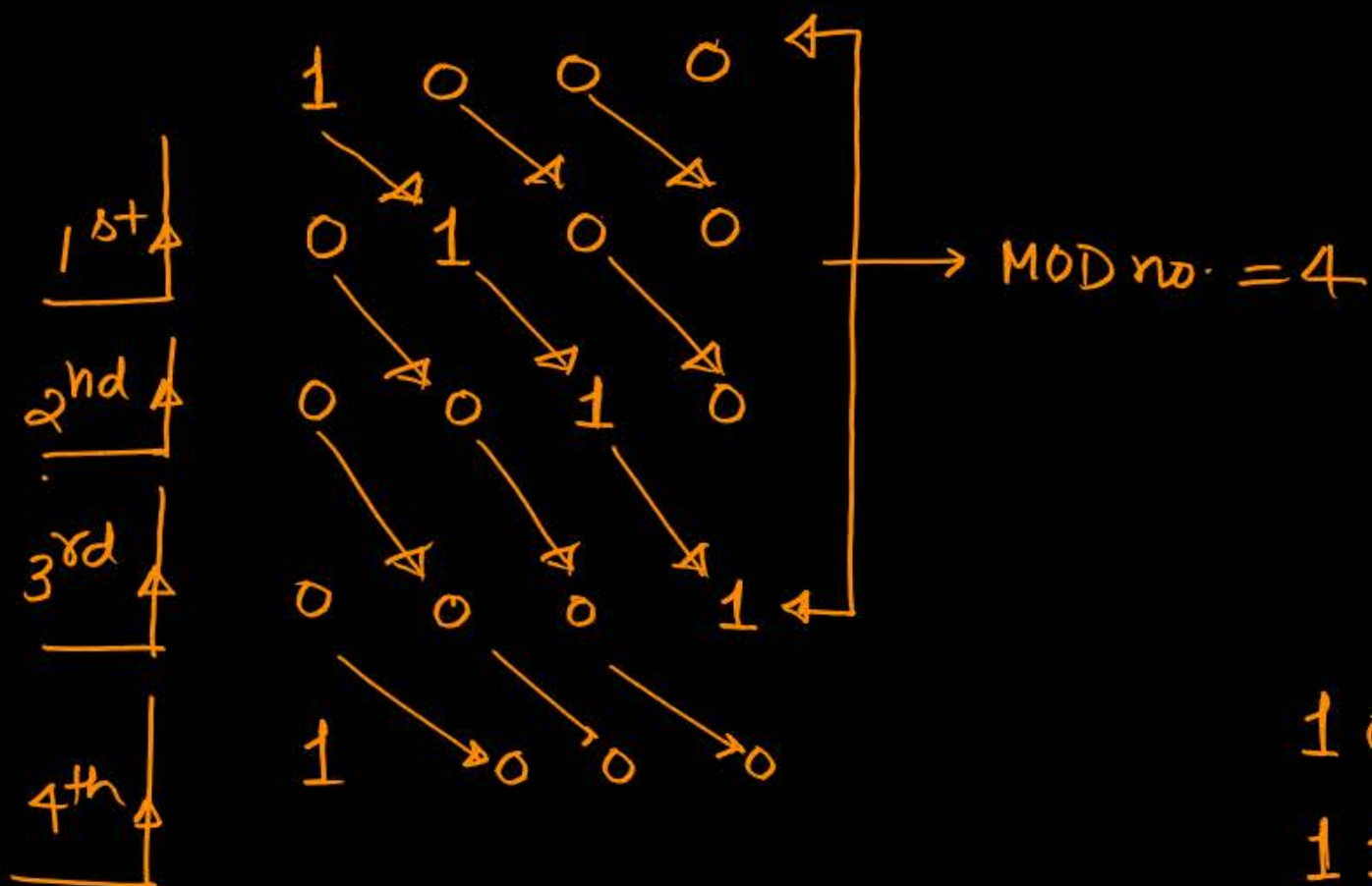
$$Q_2(n+1) = D_2 = Q_3(n)$$

$$Q_1(n+1) = D_1 = Q_2(n)$$



Working :

$f_{clk}/4$ $f_{clk}/4$ $f_{clk}/4$ $f_{clk}/4$
 Q_3 Q_2 Q_1 Q_0



at every $Q \rightarrow$ $T_{off} = 3T_{clk}$
 $T_{ON} = 1T_{clk}$
 $T_Q = 4T_{clk}$
 $f_Q = f_{clk}/4$



n-bit ring Counter $\rightarrow$ MOD no = n

at every Q $\rightarrow$

$$T_{on} = 1 T_{clk}$$

$$T_{off} = (n-1) T_{clk}$$

$$T_Q = n T_{clk}$$

$$f_Q = \frac{f_{clk}}{n}$$

$$\% \text{ duty cycle at any o/p} = \frac{1}{n} \times 100\%$$

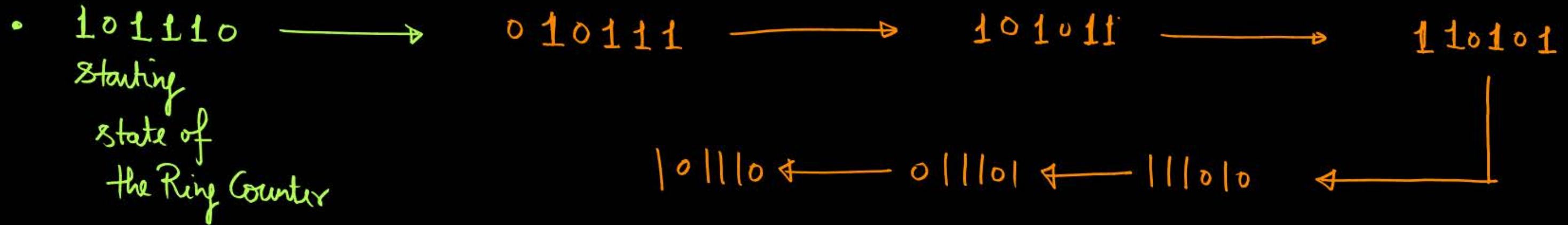
Imp points of Ring Counter : →

- With n -ff, MOD no. of the ring counter is n .
- At every O/P Q, clock frequency is divided by MOD no. n i.e. $f_Q = f_{clk}/n$
& at every O/P duty cycle $= \frac{1}{n} \times 100\%$.
- No. of unused states $= 2^n - n$



Starting state
of the Ring Counter

MOD-2 Ring Counter



MOD no. = 6

[Self Starting Ring Counter Circuit]

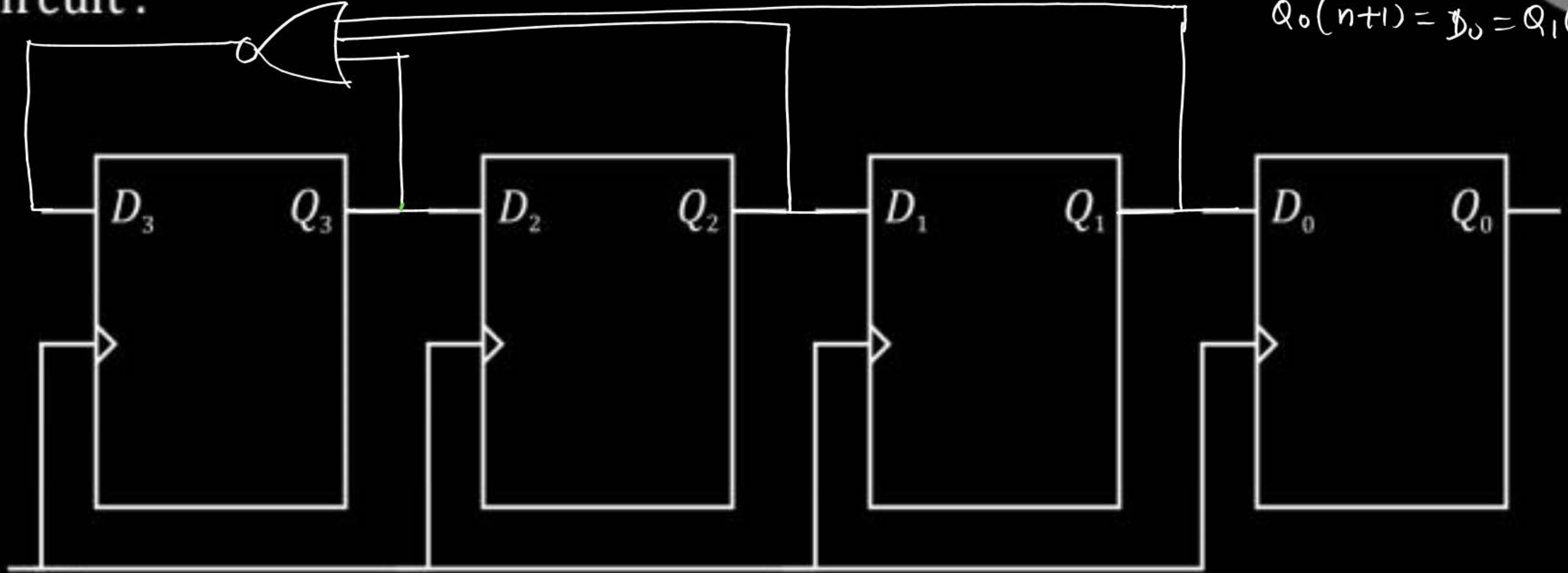


$$Q_2(n+1) = D_2 = Q_3(n)$$

$$Q_1(n+1) = D_1 = Q_2(n)$$

$$Q_0(n+1) = D_0 = Q_1(n)$$

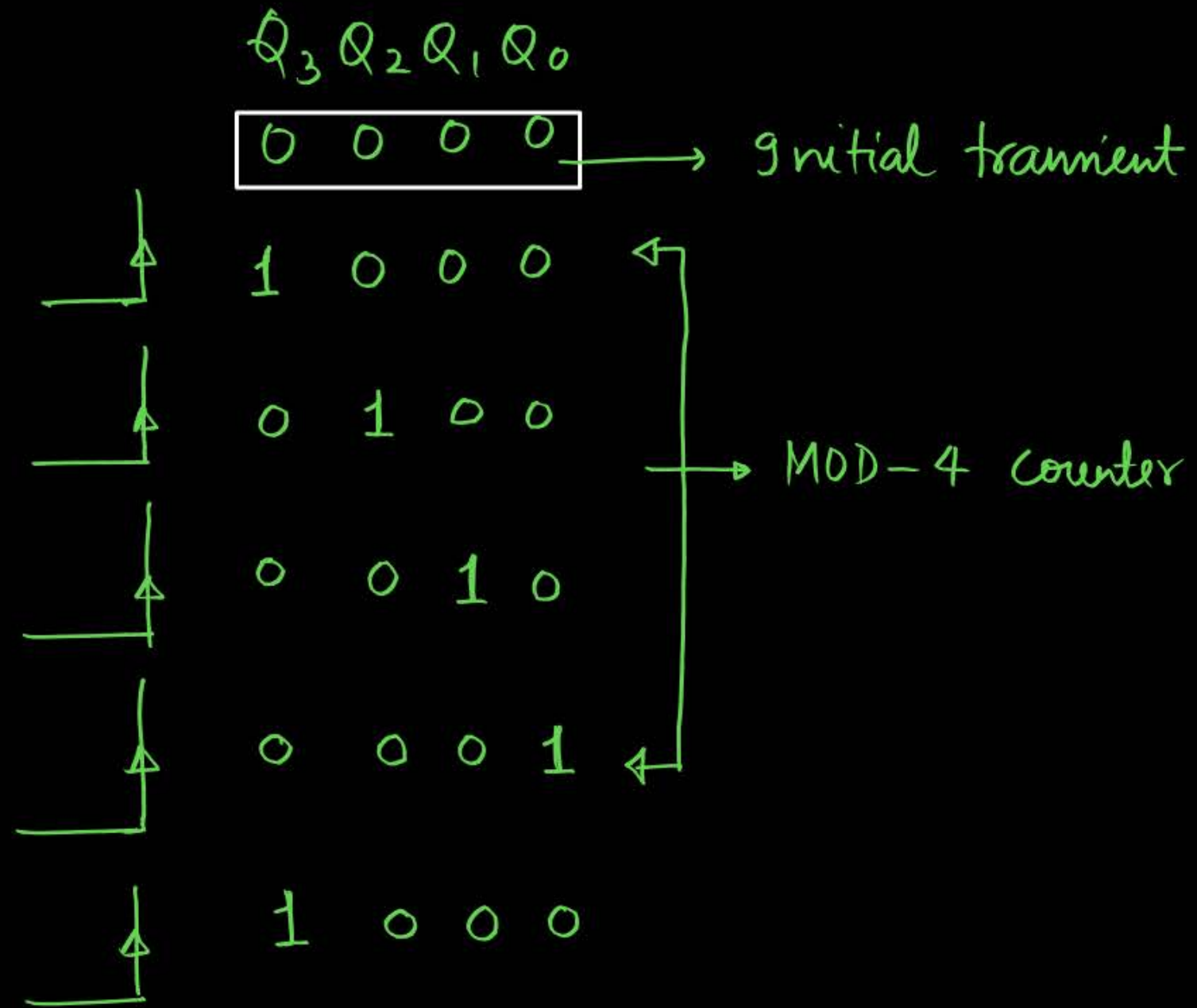
Circuit:



CLK

$$Q_3(n+1) = \overline{Q_3(n) + Q_2(n) + Q_1(n)} = D_3$$

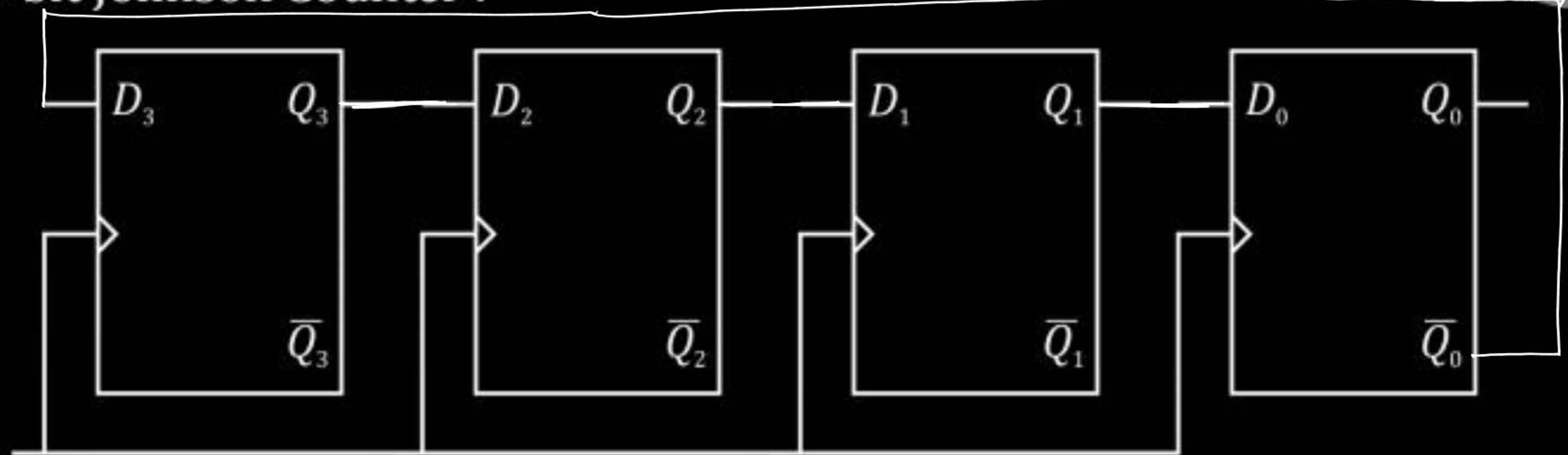
Working :



[Johnson Counter or Twisted Ring Counter]



4-bit Johnson Counter :



CLK

$$Q_2(n+1) = D_2 = Q_3(n)$$

$$Q_1(n+1) = D_1 = Q_2(n)$$

$$Q_0(n+1) = D_0 = Q_1(n)$$

$$Q_3(n+1) = D_3 = \overline{Q_0(n)}$$

Working:

	$f_{clk}/8$ Q_3	$f_{clk}/8$ Q_2	$f_{clk}/8$ Q_1	$f_{clk}/8$ Q_0
	0	0	0	0
	1	0	0	0
	1	1	0	0
	1	1	1	0
	1	1	1	1
	0	1	1	1
	0	0	1	1
	0	0	0	1
	0	0	0	0

MOD no = $2 \times 4 = 8$

at every Q

$$T_{ON} = 4T_{clk}$$

$$T_{OFF} = 4T_{clk}$$

$$T_Q = 8T_{clk}$$

$$f_Q = f_{clk}/8$$

% duty cycle = 50%

$f_{clk}/6$ Q_2	$f_{clk}/6$ Q_1	$f_{clk}/6$ Q_0
0	0	0
1	0	0
1	1	0
1	1	1
0	1	1
0	0	1
0	0	0

MOD = 6



$Q_2 Q_1 Q_0$
1 0 1
0 1 0
1 0 1

MOD-2 Counter.



Imp points of Johnson Counter : →

- With n -ff → the MOD no. of Johnson counter is $2n$.
- At every ff o/p, input clock frequency is divided by MOD no. $2n$ i.e $f_Q = f_{clk}/2n$.
& at every o/p duty cycle of the waveform will be 50% [independent of no. of ffs].
- No. of unused states $= (2^n - 2n)$

[Question]

We have 4-bit Johnson counter. Its starting state is $(1011)_2$. Then it will be in what state after 22 CLK pulses $(13)_{10}$.

Q_3	Q_2	Q_1	Q_0
1	0	1	1
0	1	0	1
0	0	1	0
1	0	0	1
0	1	0	0
1	0	1	0
1	1	0	1
0	1	1	0
1	0	1	1

→ MOD No. = 8

$$(1011)_2 \xrightarrow{2 \times 8} (1011)_2 \xrightarrow{6 \text{ CLK}} (1101)_2 = (13)_{10}$$

[Question]

We have 3-bit Johnson counter whose starting state is $(010)_2$. If input clock frequency 12 MHz then value of output frequency will be

(a) ✓ 6 MHz

(b) 2 MHz ✗

(c) 4 MHz

(d) 8 MHz

$$f_{clk} = 12 \text{ MHz}$$

$$f_{out} = \frac{f_{clk}}{2} = 6 \text{ MHz}$$

$Q_2 Q_1 Q_0$

0	1	0	$\left. \begin{array}{c} \leftarrow \\ \rightarrow \end{array} \right\} \text{MOD}-2$
1	0	1	
0	1	0	

[**Question**]

MSQ.



Which of the following is/are true ?

- (a) ✓ In ring counter duty cycle of the output waveform decreases with increase in the no. of FFs. $\frac{1}{n} \times 100\%$
- (b) Johnson counter has duty cycle of $1/2n$, where n is no. of FFs. ✗
- (c) ✓ No. of unused states in ring counter is $(2^n - n)$. ✓
- (d) No. of unused states in Johnson counter is $2^{(n-1)}$. $2^n - 2n$. ✗

[Question]



In a 3-bit ring counter with starting state $Q_2Q_1Q_0 = (110)_2$, then duty cycle of the waveform generated at Q_1 output 66.67%.

$Q_2Q_1Q_0$

1 1 0

0 1 1

1 0 1

1 1 0

MOD no. = 3

$$Q_1 \rightarrow T_{ON} = 2T_{CLK}, T_{OFF} = 1T_{CLK}$$

$$T_{Q_1} = 3T_{CLK}$$

$$f_{Q_1} = f_{CLK}/3$$

$$\begin{aligned} \% \text{ duty cycle} &= \frac{2T_{CLK}}{3T_{CLK}} \times 100 \\ &= 66.67\% \end{aligned}$$

[Shift Register]



For what purpose shift registers are used ?

- To store data
- Shifting operation is involve in IN & OUT operation.

- Types of shift registers :

→ SISO

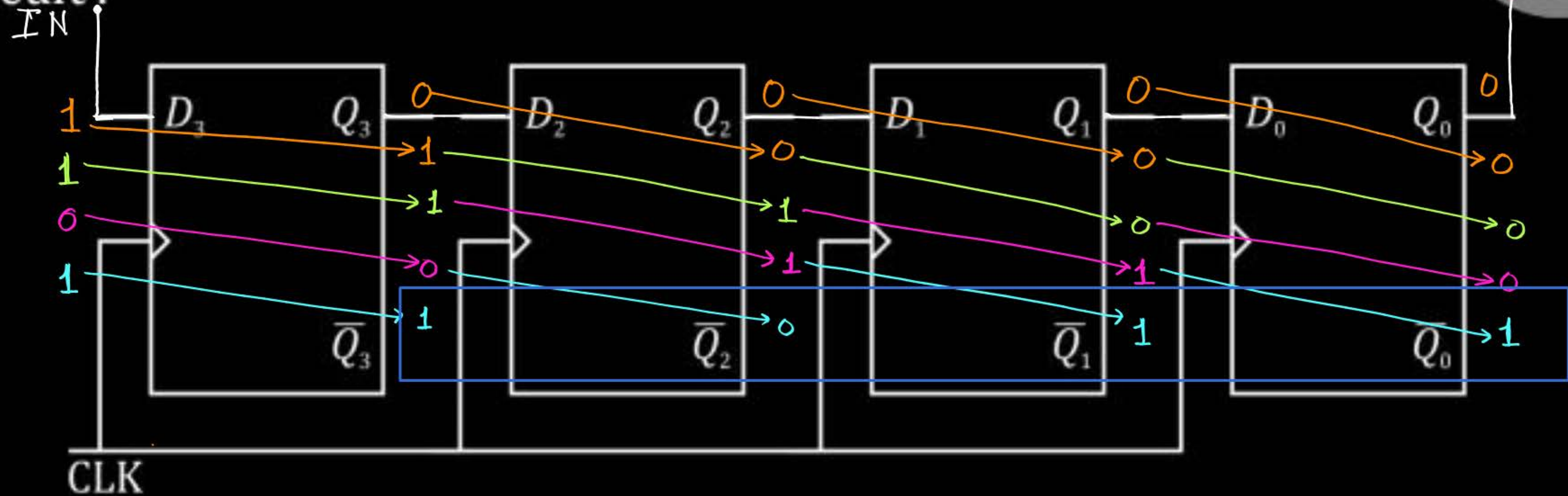
→ SIPO

→ PISO

→ PIPO

1. **SISO** : 4 bit 1 0 1 1

Circuit:

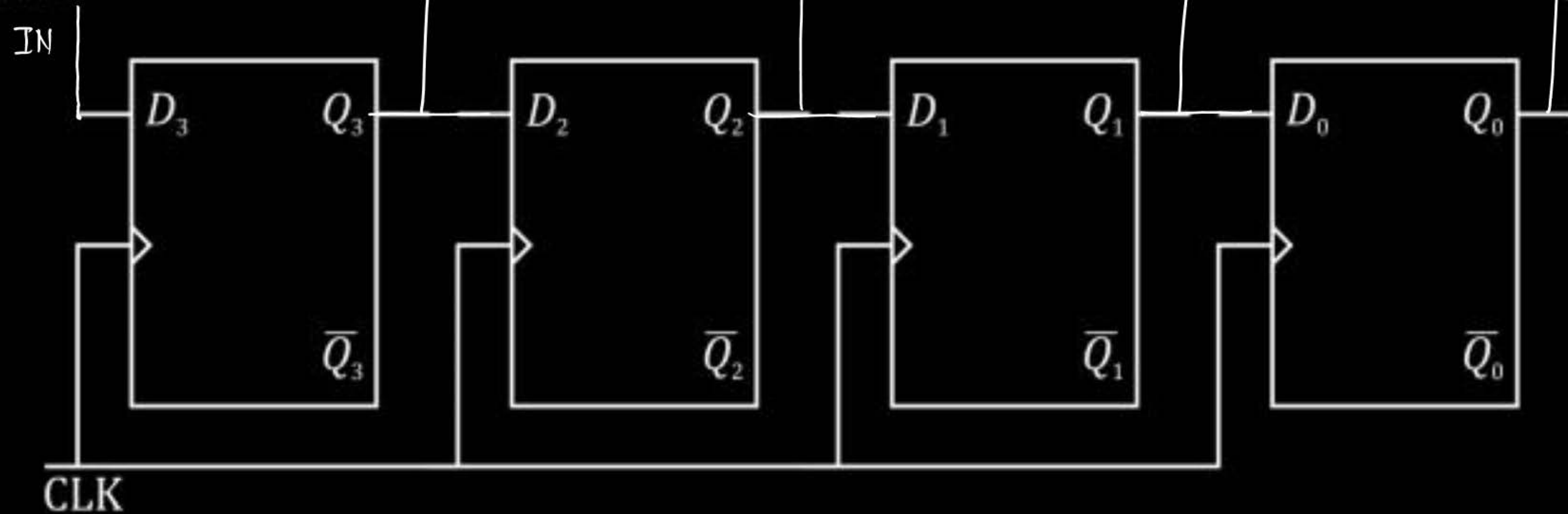


IMP Points :

- To store n -bit data serially $\rightarrow$ we require n -clocks $[nT_{clk}]$
(IN)
- To access n -bit data serially $\rightarrow$ we require $(n-1)$ clocks $[(n-1)T_{clk}]$.
(OUT)

2. SIPO :

Circuit :

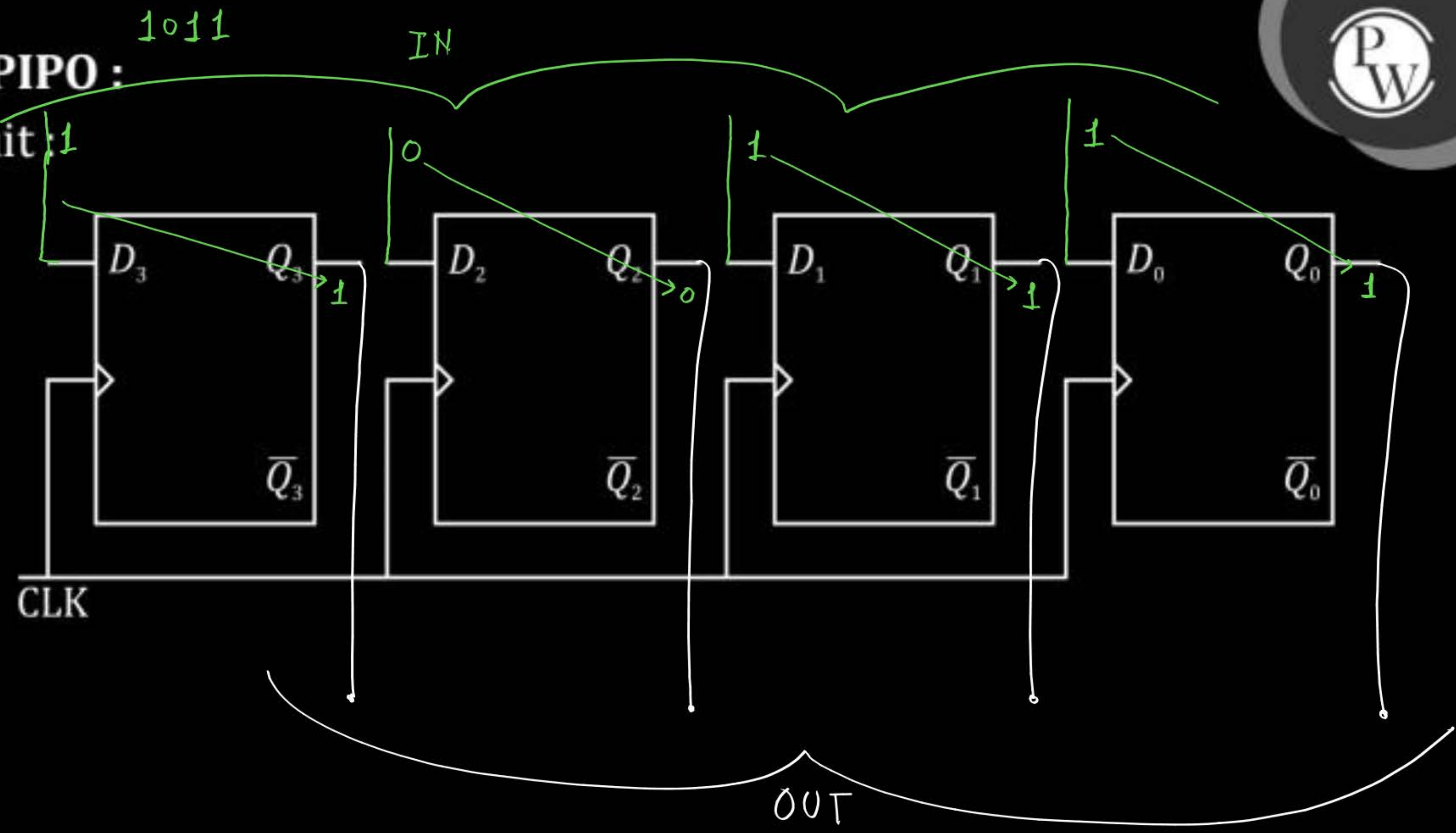


IMP Points :

- To store n -bit serially, we require $\longrightarrow n \text{ clocks} [n \cdot T_{clk}]$
(IN)
- To access n -bit parallelly, we require $\longrightarrow 1 \text{ clock} [1 \cdot T_{clk}]$.
(OUT)

3. PIPO :

Circuit :

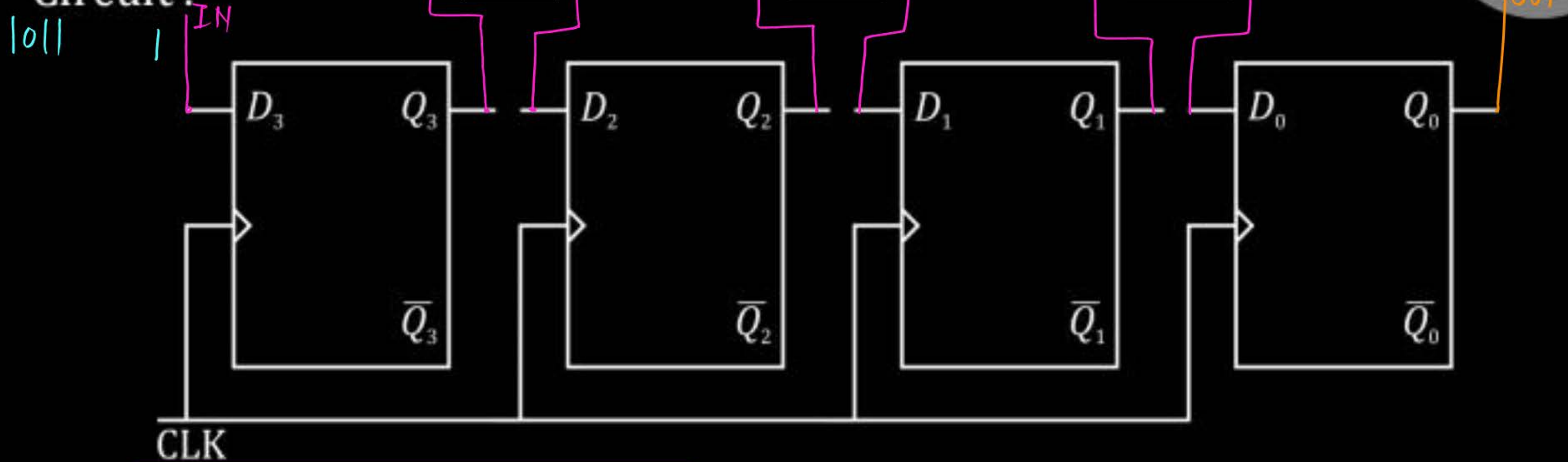


IMP Points :

- To store n -bit parallelly $\longrightarrow$ we require 1-Clock [$1T_{clk}$].
(IN)
- To access n -bit parallelly $\longrightarrow$ we require 0-Clock [$0T_{clk}$].
(OUT)

4. PISO :

Circuit :



Control = 0 $\rightarrow$ to store data
Control = 1 $\rightarrow$ to access data

IMP Points :

- To store n -bit parallelly , we require $\longrightarrow 1 \text{ Clock } [1-T_{ck}]$
- To access n -bit serially , we require $\longrightarrow (n-1) \text{ Clocks } [(n-1) T_{ck}]$.

Register	IN	OUT
SISO	$n - T_{clk}$	$(n-1) T_{clk}$
SIPD	$n - T_{clk}$	$0 T_{clk}$
PI SO	$1 - T_{clk}$	$(n-1) T_{clk}$
PIPO	$1 - T_{clk}$	$0 - T_{clk}$



Topic : 2 Min Summary

- Ring Counter
- Johnson Counter
- Shift register

Thank you

GW
Soldiers !

